CS 258 — TCP Sliding Window Simulation Report

Group Information:

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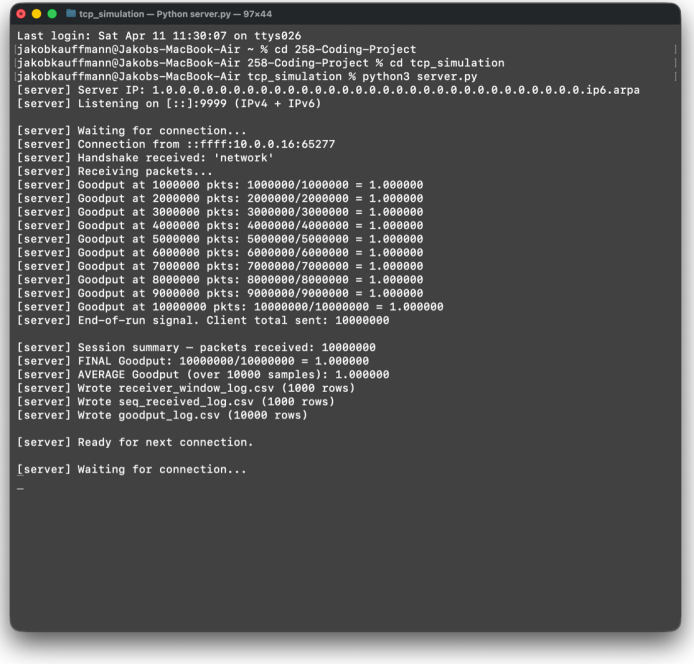
Screenshots of the Outputs

`server.py`:

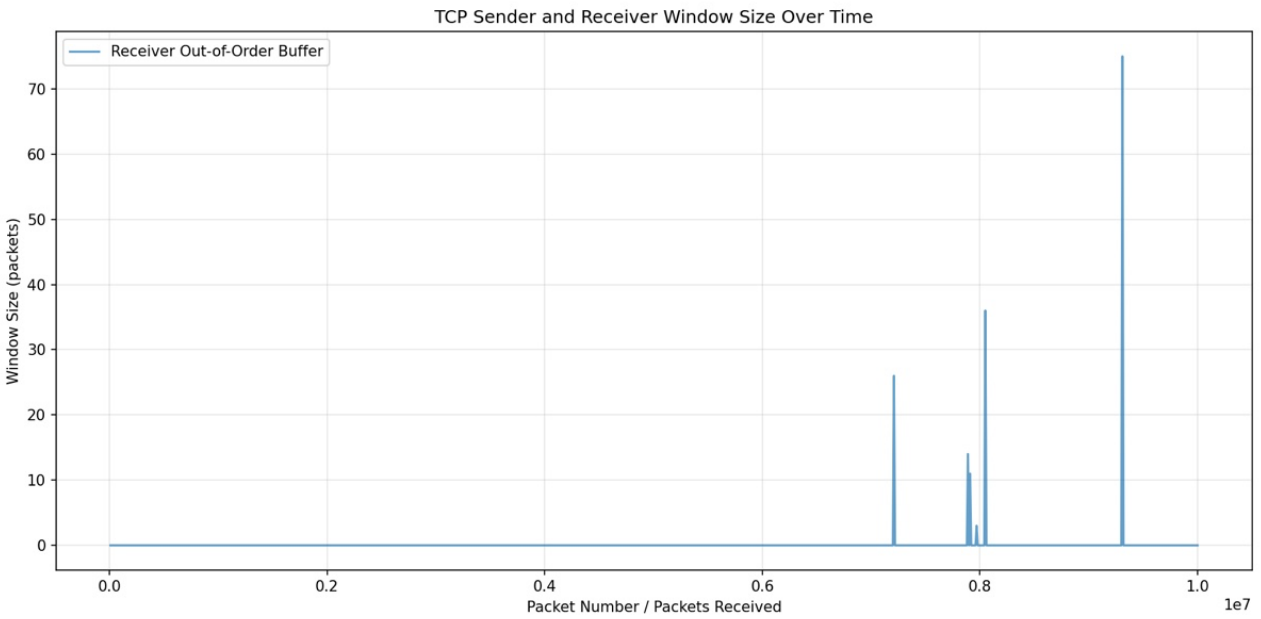

tcp_simulation — Python server.py — 97×44
Last login: Sat Apr 11 11:30:07 on ttys026
jakobkauffmann@Jakobs-MacBook-Air ~ % cd 258-Coding-Project
jakobkauffmann@Jakobs-MacBook-Air 258-Coding-Project % cd tcp_simulation
jakobkauffmann@Jakobs-MacBook-Air tcp_simulation % python3 server.py
[server] Server IP: 1.0.0.0

`client.py`:

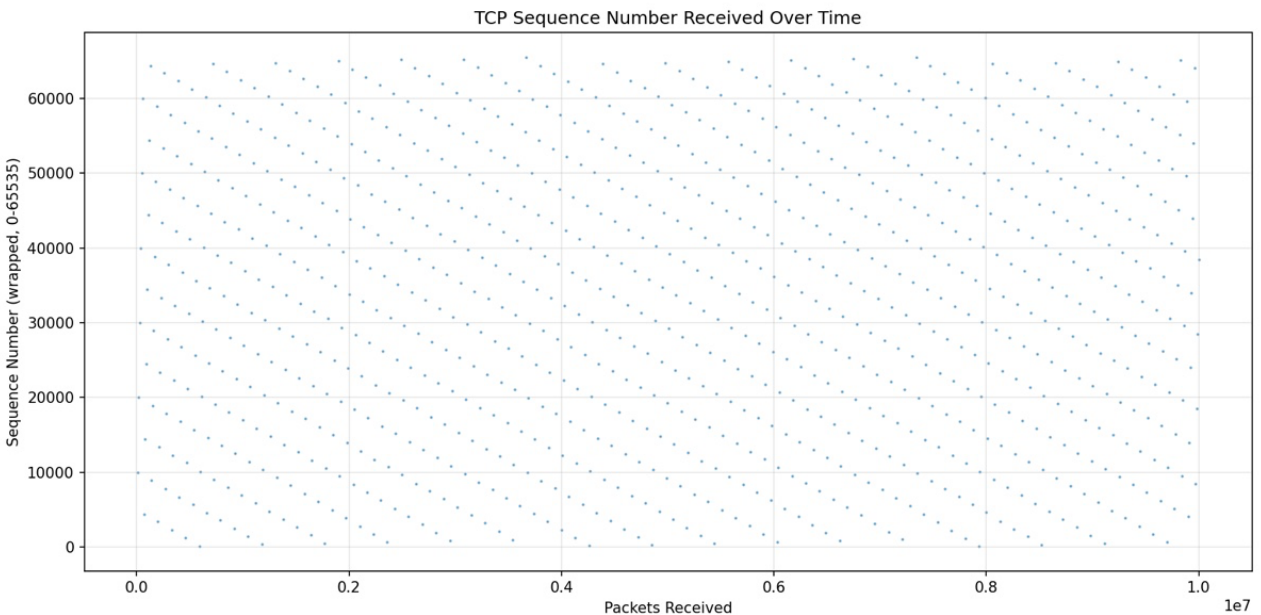
```
tcp_simulation — -zsh — 80×37
Last login: Tue Apr 14 12:45:50 on ttys001
Jakob@Shouryas-MacBook-Pro 258-Coding-Project % cd tcp_simulation
Jakob@Shouryas-MacBook-Pro tcp_simulation % python3 client.py 10.0.0.245
[client] Client IP: 10.0.0.16
[client] Connected to Server IP: 10.0.0.245, Port: 9999
[client] Server response: 'success'
[client] Sender done. Total packets sent (including retransmits): 10000000
[client] Total unique packets intended: 10000000
[client] Sent end-of-run signal.
[receiver] Server closed connection.
[client] Wrote sender_window_log.csv (1000 rows)
[client] Wrote dropped_packets.csv (100121 rows)
[client] Wrote retransmit_table.csv (100121 rows)

========= Retransmission Table =========
  # of retransmissions |   # of packets
----------------------------------------
                    1 |          99162
                    2 |            950
                    3 |              9
                   4+ |              0

Total packets initially dropped: 100121
========================================

[client] Done.
Jakob@Shouryas-MacBook-Pro tcp_simulation %
```

Graphs

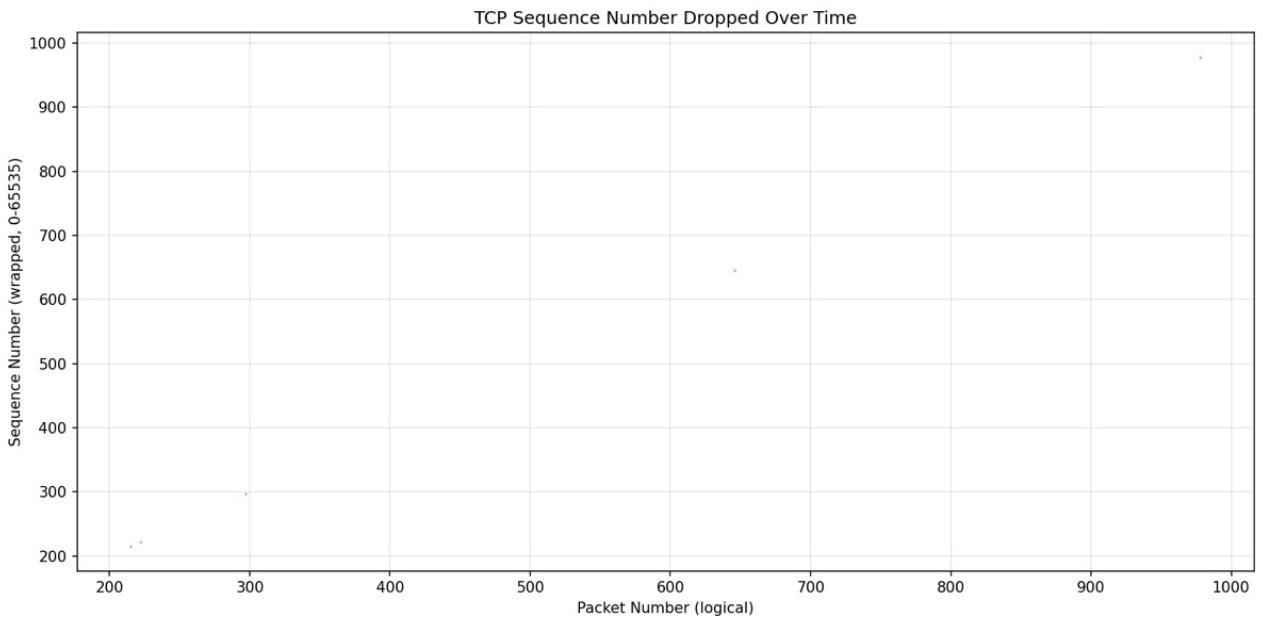
TCP Sender and Receiver Window Size Over Time



TCP Sequence Number Received Over Time



TCP Sequence Number Dropped Over Time



Goodput Over Time

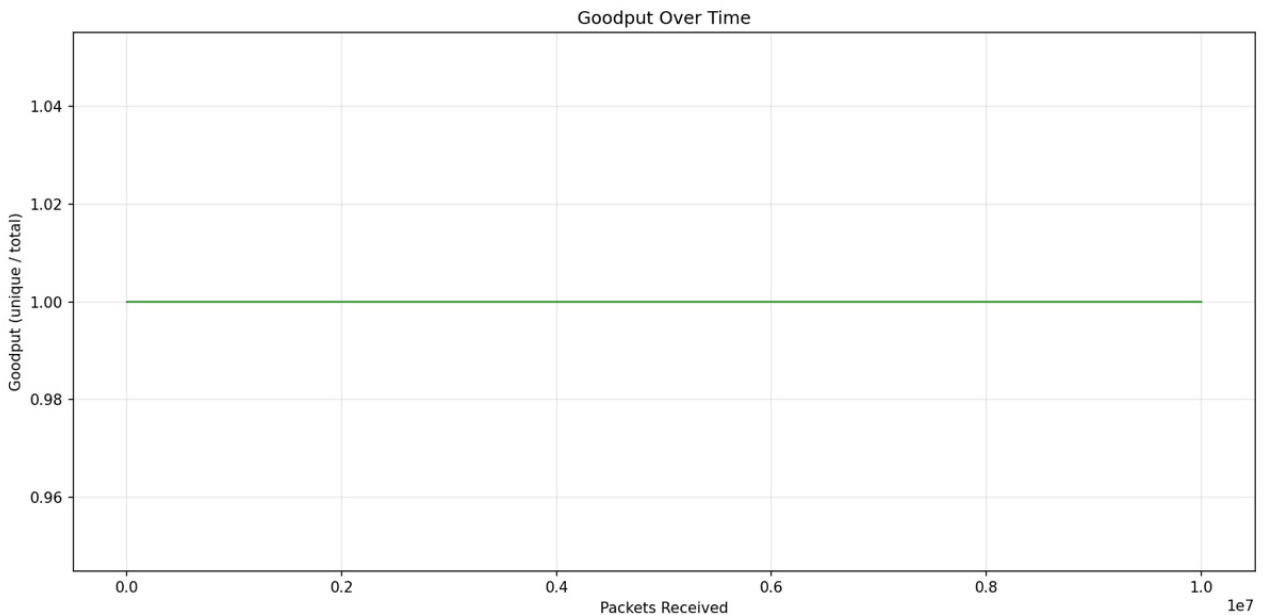


Table — Number of Retransmissions

# of Retransmissions	# of Packets
1	5
2	0
3	0
4	0